WIRESHARK NETWORK TRAFFIC ANALYSIS

Final Investigation Report

Case ID	CB-002 — Fictional training records
Tool	Wireshark
Environment	Kali Linux
Capture file	week2-traffic.pcap
Analysis focus	DNS, TCP, HTTP, and TLS traffic involving host 10.20.0.24

Purpose: Document observed network activity, distinguish evidence from inference, and recommend proportionate follow-up checks.

IMPORTANT: This is a training exercise using a generated capture. The observations do not, by themselves, prove malware execution, system compromise, or data theft.

Executive Summary

The supplied capture contains 13 packets associated with the filter ip.addr == 10.20.0.24. The host performs a DNS lookup for the approved training portal, then queries the unlisted name billing-review.example. It connects to 198.51.100.80 over TCP port 80, requests /review using HTTP, and receives an HTTP 200 OK response. A separate TCP connection is established to 192.0.2.80 on port 443, followed by a TLS application-data record.

The unlisted domain and cleartext HTTP exchange merit validation against the approved-domain list and expected business activity. However, the capture is synthetic and does not establish that the domain is malicious, that the endpoint was compromised, or that information was stolen.

Assessment area	Observation	Interpretation
DNS	Queries for portal.cedarbridge.example and billing-review.example.	Second name is not in the supplied approved-domain list; validate ownership and business purpose.
HTTP	GET /review to billing-review.example; response is 200 OK.	Shows a successful HTTP exchange, not whether the site is safe or malicious.
TLS	TCP connection to 192.0.2.80:443 followed by TLS application data.	Payload is not readable from the supplied evidence; no decryption keys are provided.

Overall	13 packets in a generated training capture.	Limited, controlled evidence; no proof of compromise, malware, or exfiltration.

Scope and Method

The review focused on the supplied generated packet capture and Wireshark screenshots. The display filter used was ip.addr == 10.20.0.24. The analysis inspected DNS queries and responses, TCP connection establishment, HTTP request/response fields, and the TLS application-data record. Findings are limited to what is visible in this capture; no external reputation lookups, endpoint telemetry, or additional logs are claimed in this report.

Packet Timeline and Evidence

Frame(s)	Observed activity	Evidence-based interpretation
1–2	Host 10.20.0.24 queries DNS for portal.cedarbridge.example; response returns 192.0.2.80.	Lookup and response for the approved training portal.
3–4	Host queries DNS for billing-review.example; response returns 198.51.100.80.	Domain is not listed in the supplied approved-domain list. This is a review point, not proof of maliciousness.
5–7	TCP handshake is established with 198.51.100.80:80.	Host establishes a TCP connection to the resolved address on HTTP port 80.
8	HTTP request: GET /review; Host header billing-review.example.	Request path and hostname are visible in cleartext HTTP.
9	HTTP response: 200 OK; content type text/plain; response body described as training page only.	Server returned a successful response. HTTP 200 does not indicate trustworthiness.
10–12	TCP handshake to 192.0.2.80:443.	A separate connection is established to port 443.
13	TLS application-data record from 10.20.0.24 to 192.0.2.80.	Application data is not readable in the supplied view; no TLS keys are available.

Key Findings

Finding 1 — Unlisted DNS name: The endpoint resolves billing-review.example to 198.51.100.80. Confirm whether the domain is expected and authorized in this training scenario.

Finding 2 — Cleartext HTTP exchange: The endpoint sends an HTTP GET request to /review and receives 200 OK. Because HTTP is unencrypted, request metadata and content are visible in the capture.

Finding 3 — Separate TLS activity: The endpoint sends TLS application data to 192.0.2.80:443. The supplied evidence does not reveal the encrypted application content.

Finding 4 — Evidence limitations: The capture is generated for training and is small. It cannot establish user intent, site reputation, endpoint compromise, malware execution, or data theft.

Recommendations

- Validate the domain: Compare billing-review.example with the approved-domain inventory and confirm the business owner and purpose of the connection.
- Review surrounding telemetry: If this were a real investigation, correlate the time window with DNS resolver logs, proxy/firewall logs, endpoint telemetry, and relevant authentication or application logs.
- Inspect HTTP context: Review the full request and response details in the supplied capture and determine whether the cleartext HTTP exchange is expected for the training environment.
- Check the TLS session: If more evidence is available, review the TLS handshake, certificate details, destination ownership, and related network logs. Do not infer the encrypted payload's contents from this record alone.
- Preserve evidence: Keep the original capture and screenshots unchanged, record the analysis filter and observations, and maintain a clear evidence trail.
- Avoid premature response actions: Do not block the domain, disable accounts, or declare an incident solely from this capture. Escalate only after corroborating evidence and follow the applicable approval process.

Limitations

- The capture is described as generated training data, not a live production incident.
- The packet set contains only 13 packets for the filtered host and provides limited context.
- No TLS decryption keys are supplied; TLS application data is not readable in the provided evidence.
- The capture alone does not prove malicious activity, compromise, malware execution, or data exfiltration.
- No independent domain reputation, endpoint investigation, or external threat-intelligence validation was performed for this report.

Lessons Learned

This investigation provided practical experience in analysing network traffic using Wireshark. The following lessons were learned:

- **Traffic filtering: Display filters such as ip.addr == 10.20.0.24 help isolate packets associated with a specific host.**
- **Protocol analysis: DNS, TCP, HTTP, and TLS traffic can be examined to understand network communication and connection behaviour.**
- **Identifying suspicious activity: An unlisted domain, such as billing-review.example, should be investigated to determine whether the connection is authorised and expected.**
- **Evidence-based investigation: A successful HTTP response or an unfamiliar domain does not automatically indicate malicious activity or a compromised system.**
- **Importance of additional evidence: DNS logs, firewall logs, endpoint telemetry, and other relevant records can provide further context during an investigation.**
- **Understanding encrypted traffic: TLS application data cannot be interpreted from the available packet view alone when the necessary decryption keys are unavailable.**

The exercise reinforced the importance of careful packet analysis, accurate documentation, and evidence-based conclusions when investigating potential network security concerns.

Conclusion

The capture documents DNS lookups, TCP connection establishment, an HTTP request/response exchange with an unlisted training domain, and a separate TLS application-data record. The most actionable observation is the request to billing-review.example, which should be validated against the approved-domain list and expected activity. The correct conclusion is that the activity warrants contextual review, not that a compromise has been confirmed.

Appendix A — Screenshot Evidence

The following screenshots correspond to the supplied Wireshark evidence: filtered packet list, HTTP GET request (frame 8), HTTP response (frame 9), and TLS application-data record (frame 13).

Figure 1. Wireshark display filter and packet list for host 10.20.0.24.

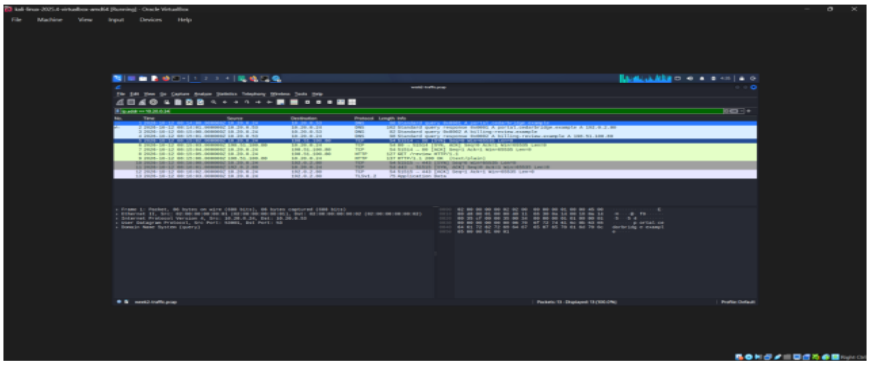


Figure 1. Wireshark display filter and packet list for host 10.20.0.24.

Figure 2. Frame 8 showing HTTP GET /review and Host: billing-review.example.

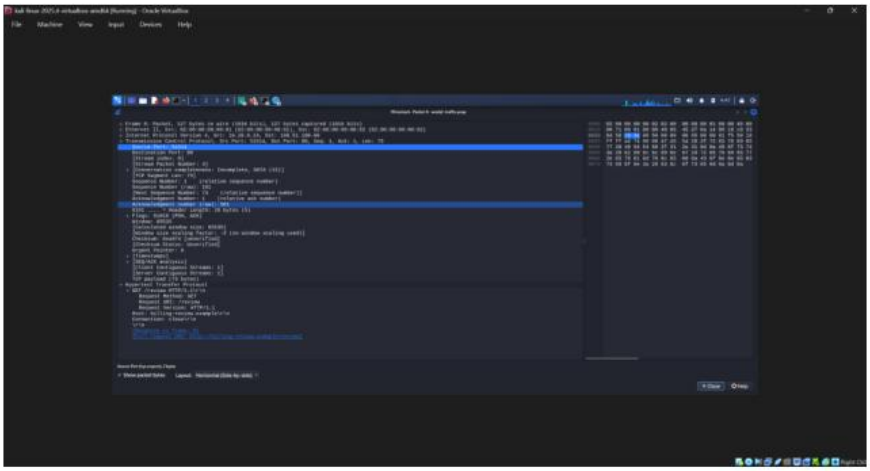


Figure 2. Frame 8 showing HTTP GET /review and Host: billing-review.example.

Figure 3. Frame 9 showing the HTTP 200 OK response.

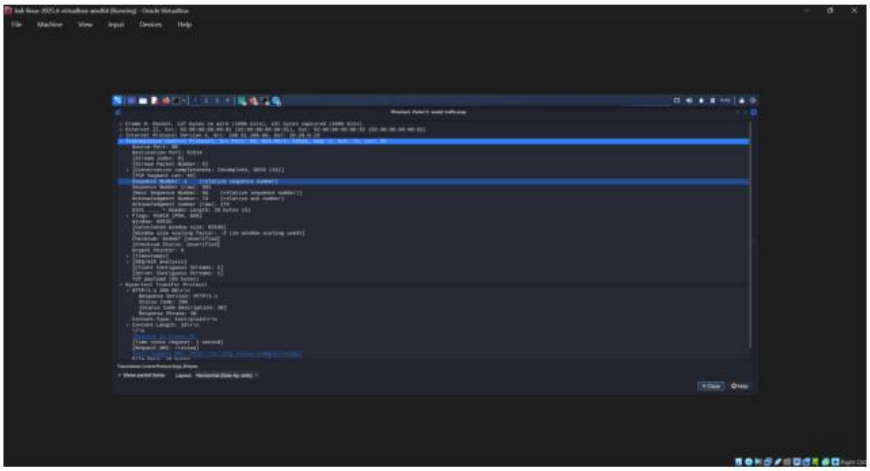


Figure 3. Frame 9 showing the HTTP 200 OK response.

Figure 4. Frame 13 showing TLS application data; payload content is not readable in the supplied evidence.

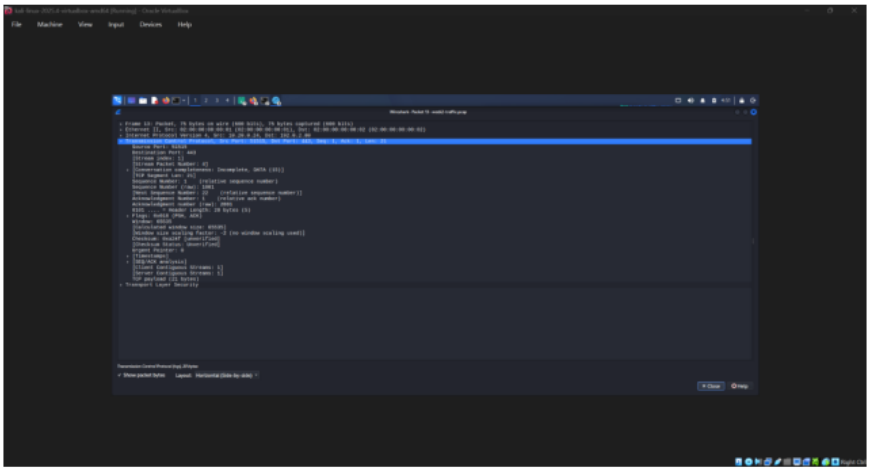


Figure 4. Frame 13 showing TLS application data; payload content is not readable in the supplied evidence.